Arrow function

```
<html>
<body>
<script>
function product()
{
let a,b;
a=parseInt(f1.no1.value);
b=parseInt(f1.no2.value);
let mul=(a,b)=>a*b;//definition
let result=mul(a,b);//call
f1.prod.value =result;
//document.writeln("The product is"+result);
}
</script>
<center>
<h3> ES6 Arrow Function</h3> <br>
  <form name="f1" >
<br>
 enter first number: <input type="text" name="no1">
  <br>
<br>
 enter second number :<input type="text" name="no2">
<br>
<br>
product is :<input type="text" name="prod">
<br>
<br>
<input type="button" value="SUBMIT" onclick="product()">
 <input type="button" value="RESET" onclick="reset()">
 </form>
```

```
</center>
</body>
</html>
```

Callback function

```
<!DOCTYPE html>
<html>
<body>
<h2> Callback function is called when we wanted </h2>
<p>Wait 3 seconds (3000 milliseconds) to c the outout</p>
<h1 id="demo"></h1>
<script>
setTimeout(myFunction, 3000);
function myFunction() {
  document.getElementById("demo").innerHTML = "welcome students !!";
}
</script>
</body>
</html>
```

Asynchronous

```
<!DOCTYPE html>
<html>
<body>
<h1>asynchronous behavior demo</h1>
<script>
console.log("First");
// An asynchronous task using set timeout
setTimeout(() => {
  console.log("Second");
```

```
}, 1000); // The delay is 1 second
console.log("Third");
</script>
</body>
</html>
```

Synchronous

```
<!DOCTYPE html>
<html>
<body>
<h1>synchronous behavior demo</h1>
<script>
console.log("First");
function longRunningTask()
{
  for (let i = 0; i < 1000000000; i++)
{
  //time consuming code
  }
  console.log("Second");
}
longRunningTask();
console.log("Third");
</script>
</body>
</html>
```

Check whether a number is even-odd using promise

```
<html>
<body>
<script>
function evenodd(num)
{
return new Promise((resolve,reject)=>
{
if (!Number.isInteger(num))
{
reject("invalid input: not an integer");
return;
}
if(num%2==0)
{
resolve(`${num} is even`);
}
else
{
reject(`${num}is odd`);
}
});
}

evenodd(4)
  .then(result => {
document.write("<br>sucess:",result);    })
  .catch(error => {
  document.write("<br>Error:",error);      });
evenodd(7)
  .then(result => {
```

```
document.write("<br>sucess:",result);
  })
  .catch(error => {
document.write("<br>Error:",error);
  });
evenodd(3.14)
  .then(result => {
document.write("<br>sucess:",result);
  })
  .catch(error => {
document.write("<br>Error:",error);
  });
</script>
</body>
</html>
```

Factorial of a given number using promise

```
<!DOCTYPE html>
<html>
<body>
    <h2>Calculate Factorial</h2>
    <input type="number" id="numberInput" placeholder="Enter a number">
    <button onclick="handleCalculate()">Calculate</button>
    <p id="result"></p>
<script>
const calculateFactorial = (n) => {
    return new Promise((resolve, reject) => {
        if (n < 0) {
            reject("Factorial is not defined for negative numbers.");
        }
```

```
        const factorial = (num) =>
          {
         if (num === 0 || num === 1) return 1;
         return num * factorial(num - 1);
       };


       resolve(factorial(n));
   });
};
async function  handleCalculate() {
   const inputField = document.getElementById("numberInput");
 const resultElement = document.getElementById("result");
try
{
   const n = parseInt(inputField.value);
const result =await calculateFactorial(n);
 resultElement.innerText = `Factorial of ${n} is: ${result}`;
       resultElement.style.color = "green";
}
catch(error)
{
         resultElement.innerText = `Error: ${error}`;
       resultElement.style.color = "red";
   }
}
</script>
</body>
</html>
```